

1. Confidence on a common scale

1. Score, prediction, and normalized margin

For binary $h_t(x) \in \{-1, +1\}$, AdaBoost's real score and prediction are

$$F_T(x) = \sum_{t=1}^T \alpha_t h_t(x), \quad \text{sign}(F_T(x)).$$

For a labeled example (x, y) and nonzero coefficient vector α , the **normalized functional margin** is

$$\rho(x, y) = \frac{y F_T(x)}{\|\alpha\|_1}, \quad \|\alpha\|_1 = \sum_{t=1}^T |\alpha_t|.$$

Its sign records correctness and its magnitude records signed-score confidence after removing arbitrary positive coefficient scale. Under $\alpha \rightarrow c\alpha$ with $c > 0$, the raw score scales by c , while ρ and the prediction do not change.

2. Do not confuse margin with distance

The weak-hypothesis-output vector is $h(x) = (h_1(x), \dots, h_T(x)) \in \mathbb{R}^T$.

Its unsigned L^∞ distance to the α -normal hyperplane is

$$d_\infty(h(x), \{z : \alpha \cdot z = 0\}) = \frac{|\alpha \cdot h(x)|}{\|\alpha\|_1} = \frac{|F_T(x)|}{\|\alpha\|_1}.$$

This distance is measured in weak-hypothesis-output space, not in the original input space and not between coefficient vectors. On a correctly classified example it equals $\rho(x, y)$; on a misclassified example the distance stays positive while $\rho(x, y) < 0$.

3. When normalization gives a convex combination

If every $\alpha_t \geq 0$ and $\sum_t \alpha_t > 0$, then

$$\bar{F}(x) = \frac{F_T(x)}{\|\alpha\|_1} = \sum_{t=1}^T \frac{\alpha_t}{\|\alpha\|_1} h_t(x)$$

is a convex combination: its weights are nonnegative and sum exactly to one. Division by the positive norm preserves the sign. If signed coefficients are allowed, this convex-hull claim needs sign closure of the base class or an explicit reparameterization. The L^1 denominator alone does not make negative coefficients convex weights.

2. Margin distributions and population control

1. Empirical cumulative low-margin fraction

For an i.i.d. sample $S = ((x_i, y_i))_{i=1}^m$ and threshold θ ,

$$\hat{R}_{S, \theta}(\bar{F}) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}[y_i \bar{F}(x_i) \leq \theta].$$

For a nonzero positive-coefficient AdaBoost score normalized by its L^1 coefficient norm, $y_i \bar{F}(x_i) = \rho(x_i, y_i)$. As θ increases, this empirical margin distribution is nondecreasing: $\theta_1 \leq \theta_2$ implies $\hat{R}_{S, \theta_1} \leq \hat{R}_{S, \theta_2}$. At zero, apart from the chosen zero-score tie convention, it recovers empirical sign error. Positive thresholds additionally count correctly classified examples with low normalized confidence.

A right-shifted training-margin curve is training evidence only. It is not a test-error curve and does not force future test error to fall.

2. Normalized convex ensemble class

For a binary base class $\mathcal{H}$,

$$\text{conv}(\mathcal{H}) = \left\{ \sum_{j=1}^p \mu_j h_j : p \geq 1, \mu_j \geq 0, h_j \in \mathcal{H}, \sum_{j=1}^p \mu_j = 1 \right\}.$$

The zero function is in this class only if it can actually be represented by such a convex combination; it is not admitted automatically. An arbitrary real linear combination is likewise not automatically a member. A positive-coefficient AdaBoost score enters after division by its positive L^1 coefficient norm. For this standard convex hull,

$$\hat{\mathfrak{R}}_S(\text{conv}(\mathcal{H})) = \hat{\mathfrak{R}}_S(\mathcal{H}).$$

The equality follows because a linear objective over convex weights is maximized at an extreme point. It is a normalized-class fact, not a monotone-rounds claim.

3. Fixed-threshold boosting margin bound

Let $\mathcal{H} : \mathcal{X} \rightarrow \{-1, +1\}$ have $d = \text{VCdim}(\mathcal{H})$ with $1 \leq d \leq m$. Fix $\rho_0 > 0$ and $\delta \in (0, 1)$. For an i.i.d. sample S of size m , with probability at least $1 - \delta$, simultaneously for every $\bar{F} \in \text{conv}(\mathcal{H})$,

$$R(\bar{F}) \leq \hat{R}_{S, \rho_0}(\bar{F}) + \frac{2}{\rho_0} \sqrt{\frac{2d \log(e \frac{m}{d})}{m}} + \sqrt{\frac{\log(\frac{1}{\delta})}{2m}},$$

where $R(\bar{F}) = \Pr[y \bar{F}(x) \leq 0]$. The right side contains the empirical low-margin mass, a base-class complexity term, and a confidence term. It is an upper bound, not observed population error or an equality.

4. Read the quantifiers and tradeoff correctly

The order is: fix $\rho_0 > 0$; draw the i.i.d. sample; then, with probability at least $1 - \delta$, assert the inequality uniformly over normalized ensemble functions. The displayed event is not uniform over every threshold. Choosing a favorable threshold after seeing the sample needs the corrected all-threshold theorem with its additional log-log term. Increasing ρ_0 can increase the cumulative empirical term but decreases the $\frac{1}{\rho_0}$ complexity penalty. Increasing m shrinks both nonempirical square-root terms on an approximate $\frac{1}{\sqrt{m}}$ scale, with logarithmic dependence retained. The displayed bound has no explicit round count T , but does not imply that more rounds improve margins or test error.

3. Exponential loss as the training surrogate

1. Pointwise and empirical upper bounds

For signed score $z = yF(x)$, pointwise

$$1[z \leq 0] \leq e^{-z}.$$

If $z \leq 0$, both sides are at least one; if $z > 0$, the indicator is zero.

Averaging gives $\hat{R}_S(\text{sign}(F)) \leq m^{-1} \sum_i e^{-y_i F(x_i)}$. For $\varphi(z) = e^{-z}$,

$\varphi''(z) = e^{-z}$; the loss is infinitely differentiable and convex but not globally L -smooth on $\mathbb{R}$. On $z \in [-B, B]$ it is e^B -smooth.

Differentiability, convexity, global or bounded-domain smoothness, domination, calibration, robustness, and population performance are distinct. This empirical inequality does not certify test error.

4. One AdaBoost round as one coordinate step

1. Objective and current example distribution

Fix a finite binary dictionary $\mathcal{H} = \{h_1, \dots, h_N\}$. With $\beta \in \mathbb{R}^N$,

$$F_\beta(x) = \sum_{j=1}^N \beta_j h_j(x), \quad \mathcal{L}(\beta) = \frac{1}{m} \sum_{i=1}^m e^{-y_i F_\beta(x_i)}.$$

At round t , normalize the current exponential masses:

$$D_t(i) = \frac{e^{-y_i F_{t-1}(x_i)}}{\sum_{r=1}^m e^{-y_r F_{t-1}(x_r)}}.$$

Coordinate descent changes one coefficient while holding all others fixed. The variant used here selects a maximum-descent coordinate in this fixed dictionary, then performs exact line minimization.

2. Weighted error selects the coordinate

For coordinate k , define

$$\varepsilon_{t,k} = \sum_i D_t(i) 1[h_k(x_i) \neq y_i].$$

Let $C_t = \mathcal{L}(\beta^{t-1}) > 0$. Moving along h_k gives

$$\frac{d}{d\eta} \mathcal{L}(F_{t-1} + \eta h_k)|_{\eta=0} = C_t (2\varepsilon_{t,k} - 1).$$

The factor C_t is positive and common to all coordinates, so minimum current weighted error is exactly maximum immediate descent. It may be ignored for coordinate ranking, but is not generally equal to one and must remain in the derivative equality.

3. Exact line minimum and round factor

For a selected coordinate with weighted error ε_t , the line objective is

$$\mathcal{L}(F_{t-1} + \eta h_t) = C_t ((1 - \varepsilon_t)e^{-\eta} + \varepsilon_t e^{\eta}).$$

For $0 < \varepsilon_t < \frac{1}{2}$, strict convexity gives the unique finite minimum

$$\alpha_t = \frac{1}{2} \log \frac{1 - \varepsilon_t}{\varepsilon_t} > 0.$$

The minimum factor is $2\sqrt{\varepsilon_t(1 - \varepsilon_t)}$. Exact exponential-loss line search, not convexity alone, licenses this coefficient.

4. Conditional equivalence

The forward-greedy coordinate rule compares only positive directions $+e_k$. Its minimizing coordinates are exactly the base hypotheses with minimum current D_t -weighted error. One such step coincides with one AdaBoost round only when:

1. base hypotheses come from the fixed finite binary dictionary (or a separately defined oracle analogue);
2. the coordinate rule and weak learner use the same tie-breaking rule;
3. $0 < \varepsilon_t < \frac{1}{2}$; and
4. line search on the empirical exponential objective is exact.

Approximate weak-learner selection or line search gives only an approximate greedy step. Cyclic, random, or block-coordinate descent, and another optimizer of the same loss, need not reproduce AdaBoost's path, coefficients, classifier, margins, or generalization behavior.

Boundary audit: $\varepsilon_t = 0$ has only a $+\infty$ limiting positive step; $\varepsilon_t = \frac{1}{2}$ gives step zero; $\frac{1}{2} < \varepsilon_t < 1$ gives a finite negative step; and $\varepsilon_t = 1$ has only a $-\infty$ limiting step. Replacing a negative direction by $-h_k$ requires sign closure or an explicit equivalent reparameterization.

5. Persistent weights: what they do and do not show

Because $D_t(i)$ is proportional to $e^{-y_i F_{t-1}(x_i)}$, a persistently negative or small signed score can gain later weight and influence weak-learner selection. This identifies current training difficulty, not its cause: label noise, a rare valid region, an insufficient weak class, or early optimization can look alike. A large weight proves none of them and does not imply monotone test-error worsening. Use the update for mechanics, exact line search for objective decrease, a fixed-threshold margin theorem for a population certificate, and untouched held-out data for fitted test behavior.